

How to Prevent Cyber Attacks

Cyber security is the process of detecting and preventing unauthorised access to computer systems. This article explains how to defend against cyber attacks



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The use of computers and the Internet in industry and commerce has expanded dramatically. However, ensuring the security of network and privacy of data continues to be a big challenge. Even if the network is not connected to the Internet or other networks, it is still vulnerable to cyber attacks transported by removable media or from individuals having access to the network.

Anticipating what the attacker is going to try to penetrate the system or network is an administrator's most valuable tool. If the administrator knows where to look and detects the attacker in the middle of the attack, it

is a lot easier to safeguard the system and network. If a hacker is not caught, he can sometimes go unnoticed for months and steal large amounts of data or even destroy the entire network or system.

Physical access controls, implementation of security policies, software controls, hardware controls, encryption, etc can be used to stop cyber attacks. Typically, solutions that ensure security involve use of a combination of attack detection/identification, prevention and removal using some special techniques and tools.

Intrusion detection system

An intrusion detection system is used to monitor network resources, systems

and activities against unauthorised access. It checks and reports to administrator when it finds a possible intrusion. Usually, the intrusion detection system catches programs that are already written and have a noticeable shell code in them. (Shell code is like the hacker's own code that he wants to run when he gains control over a program.)

Intrusion detection systems are broadly classified as knowledge-based and behaviour-based systems. Most of them are knowledge-based and detect intrusion by matching signatures of already known attacks against changes in system files and firmware of network components.

Knowledge-based systems are commercially available and are very reliable for detecting known attacks, but these systems fail whenever new attacks come into picture. On the other hand, behaviour-based systems can detect a previously unreported attack because they have the processing power and the granularity to analyse attacks. These systems monitor behavioural activities on systems and network components and finally alert the administrator of possible intrusion.

Defense at switch level

In computer networks, switches are used to analyse and route packets travelling on the network towards their destination. Most of these switches have rate-limiting, traffic shaping, transmission control protocol (TCP) splicing, deep packet inspection, bogus IP filtering and access control list capability.

Here it is noteworthy that switches can detect and stop only those types of intrusions that are within their capability. For example, delayed binding or